# Adhithyan Sakthivelu

admkr.2010@gmail.com  Google Scholar  Linked-In  GitHub  +1(618)-767-4314


## Education

**The University of Texas at Austin**, TX, USA — 2025 – Present
Ph.D., Civil, Architectural and Environmental Engineering
*Selected Courses: Scientific Machine Learning, Water System Simulation and Modeling*

**Stanford University**, CA, USA — 2021 – 2023
M.S., Civil and Environmental Engineering
*Selected Courses: Electricity Markets and Economics, Convex Optimization 1 & 2, Advanced Feedback Control, Decision Making under Uncertainty*

**National Institute of Technology, Trichy**, India — 2014 – 2018
B.Tech., Instrumentation and Control Engineering
*Selected Courses: Data Structures and Algorithms, Control Systems*

## Research Experience

**Graduate Research Assistant**, WATSUP Lab, The University of Texas at Austin — 2025 – Present

- Developed a scenario-based stochastic Model Predictive Control (MPC) framework for Water Distribution Systems (WDS) to optimize Demand Response (DR) participation under uncertainty
- Formulated the WDS optimization problem that minimizes Time-of-Use (TOU) costs tariffs with DR revenues and penalties while satisfying strict operational constraints and limits

**Graduate Research Assistant**, WE3 Lab, Stanford University — 2023 – 2025

- Developed risk-aware bidding strategy for industrial facilities in DR participation under uncertainty
- Analyzed and parametrized stochastic events such as DR and Algae for energy optimization for wastewater treatement and desalination facilities
- Released a package called *dr-simulator* that models DR events based on simulation and DR program parameters
- Enhanced network-based facility optimization using MPC with wastewater treatment plant as casestudy

**Graduate Research Assistant**, SLAC National Accelerator Laboratory — 2022 – 2024

- Designed heuristic control strategies for DERs in a peer-to-peer transactive energy markets based control
- Modeled devices like batteries, PV systems, HVAC, water heaters, and EVs for market performance analysis and validated with physical devices
- Contributed to market mechanism development for transactive energy systems (IEEE PESGM Best Paper Award)

## Research Publications

### Journal Articles


[1] F. T. Chapin, A. K. Rao, **A. Sakthivelu**, C. I. Tucker, E. David, C. S. Chen, E. Musabandesu, and M. S. Mauter, “Retail electricity costs and emissions incentives are misaligned for commercial and industrial power consumers”, *arXiv preprint arXiv:2511.10775*, 2026.

[2] **A. Sakthivelu**, F. T. Chapin, J. Bolorinos, and M. S. Mauter, "Demand response event simulator and risk-aware bidding tool for industrial customers", *Applied Energy*, vol. 409, p. 127 456, 2026.

[3] E. David, **A. Sakthivelu**, A. K. Rao, and M. Mauter, "Us incentive based demand response program parameters", 2025. DOI: 10.25740/ck480bd0124.

[4] A. K. Rao, F. T. Chapin, E. Musabandesu, **A. Sakthivelu**, C. Tucker, D. Wettermark, and M. S. Mauter, "How much can we save? upper bound cost and emissions benefits from commercial and industrial load flexibility", *arXiv preprint arXiv:2511.14928*, 2025.

[5] A. Sreekumar, **A. Sakthivelu**, and L. Kiesling, "Auction theory and device bidding functions for transactive energy systems: A review", *Current Sustainable/Renewable Energy Reports*, vol. 10, no. 3, pp. 102–111, 2023.

### Conference Proceedings

[6] **A. Sakthivelu** and L. Sela, "Rethinking pump and tank design in water systems for energy flexibility", in *ASCE 2027*, In Preparation, 2027.

[7] **A. Sakthivelu** and L. Sela, "Model predictive control for reliable demand response participation in water distribution systems", in *CCWI 2026*, Submitted, 2026, pp. 1–8.

[8] A. Sreekumar, **A. Sakthivelu**, R. Baltaduonis, L. Kiesling, S. Hoedl, and D. P. Chassin, "A real-time limit order book as a market mechanism for transactive energy systems", in *2023 IEEE Power & Energy Society General Meeting (PESGM)*, IEEE, 2023, 1–5 (Best Paper Award).

## Employment History

**Power System Engineer**, Post Road Foundation, Oakland, CA — Jan – May 2025

- Developed bidding functions for "TESS" (Transactive Energy Service System) in support of DOE-funded Connected Communities project
- Contributed to transactive energy market mechanisms for distributed energy resource coordination

**Consultant**, Deloitte US India Consulting, Bengaluru, India — 2018 – 2021

- Customized Salesforce Lightning Platform and Apttus CPQ managed packages for Lead-to-Cash lifecycle
- Analyzed and processed raw data, transforming them to Salesforce-compatible schema; automated data handling
- Contributed to internal sustainability efforts via waste management optimization and awareness campaigns

**Project Intern**, Dr. Reddy's Laboratory, Hyderabad, India — May – Aug 2017

- Optimized Water Purification and Distribution Control System, achieving a 50% reduction in instrument error

## Projects

**Energy-Efficient Ride-Sharing Algorithm via Distributed Optimization** — Mar – May 2022
*Course Project: Convex Optimization II*

- Developed a ride-sharing optimization with focus on energy-efficiency and decentralized decision-making agents for rider pickup and drop-off, as part of localized clusters of connected cars
- Used distributed convex optimization (ADMM) and Google's Map API to solve several local energy-efficient drop-off optimizations on the edge (run on each vehicle) in parallel, followed by solving a master energy-efficient pickup optimization problem

**Online US Wind Energy Atlas** May – Sep 2021

*GIS website developer*

- Created a dynamic web tool for windfarm feasibility, presented at Glasgow COP26
- Developed QGIS-based visualizations and energy potential metrics
- Website: windenergyatlas.stanford.edu

## Leadership and Mentorship

**Undergraduate Summer Fellowship Mentor**, WE3 Lab, Stanford University May – Sep 2024

*Eres David, Casey Chen*

- Mentored two undergraduate researchers through SURGE and MUIR programs in developing machine-readable databases for demand response and tariff optimization
- Guided students through data collection and schema design for DR program parameters

**Energy Theme Lead**, Stanford–IIT Bombay Workshop on Sustainability Jul 2024

- Led breakout sessions on the Energy theme across a 3-day international workshop co-sponsored by the Stanford Doerr School of Sustainability and IIT Bombay
- Co-authored a white paper identifying collaborative research and educational opportunities in energy between the two institutions
- Delivered a summary presentation to faculty from both institutions synthesizing discussion outcomes and proposed research directions

## Fellowships and Awards

- **UT Austin Engineering Fellowship**, In recognition of outstanding academic record 2025 – Present
- **Best Paper Award**, IEEE Power & Energy Society General Meeting (PESGM), Orlando, Florida 2023

## Skills

- **Programming:** Python, Julia, MATLAB, Salesforce Apex, SOQL
- **Libraries:** CVXPY, Pyomo, Gurobi, Numpy, Pandas, Pytest, PyQGIS
- **Tools:** Git, GitHub Actions, LaTeX, QGIS, ArcGIS
- **Platforms:** Salesforce Lightning, Apex, MS Excel – Advanced